Python Project - Week 5

Modules and Classes - Final Integration

Today's goal

Use a module and a class to organize the functions and character information developed in Weeks 1-4.

1. Guided Coding

Study the example code below with your team. Type it into your editor and run it. Do not copy and paste.

```
class Character:
    def __init__(self, name):
        self.name = name
        self.hp = 100
        self.gold = 30

    def show(self):
        print("Name =", self.name)
        print("HP   =", self.hp)
        print("Gold =", self.gold)

hero = Character("Alex")
hero.show()
```

2. Team Coding Tasks

1. Move the reusable functions from Weeks 3-4 into game_utils.py.

2. Import game_utils in the main program.

3. Create a Character class containing name, hp, gold, attack, and inventory.

4. Connect the Character object to the battle, status, inventory, save, and load functions.

5. Run the complete program from the beginning and confirm that battle and save/load work together.

3. Expected Output / Checkpoint

```
========================================
          YONSEI QUEST
========================================
1. New game
2. Load game

[Main Menu]
1. Status
2. Forest battle
3. Shop
4. Save
5. Exit
```

4. Submission Check

• The program runs without errors.

• The program runs from beginning to end.

• Every team member can explain the code added this week.

• Save the completed source code as week5_project.py.